

KRAB domain-containing zinc-finger proteins in ancient humans

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1 Background

The Krüppel-associated box (KRAB) domain-containing zinc-finger proteins (KZFPs) form the largest family of transcriptional regulators in mammals [2]. All of the nearly 350 human members of this family share a structure which can be traced back to the coelacanth [3]. It consists of two functional domains: the KRAB domain and the C2H2 zinc finger array. The N-terminus KRAB domain mediates long term repression of transcription through recruitment of KAP1 (TRIM28) and additional epigenetic co-regulators involved in histone methylation. The variable C-terminus C2H2 zinc finger array conveys DNA sequence specificity.

Studies have shed light on the repressive activities of KZFPs towards transposable elements (TEs), building a case for an arms race for genomic stability between KZFPs and TEs. This model has recently been refined by several observations indicating that KZFPs do not merely suppress TEs subsequently to their waves of genome invasions. KZFPs rather seem to "tame" TEs into sculpting complex gene regulatory networks in a species-specific manner. Despite the existence of such strong drivers of inter-species genetic variation, biological functions are surprisingly well conserved across mammals and vertebrates. To solve this paradox, a darwinian process has been proposed: a high environmental pressure may select on the vast genetic variability caused by the activities of TEs and KZFPs thereby forcing a parallel evolution of biological functions. However, environmental pressure may be relieved from certain organs (e.g the brain) through increased social interactions (e.g protection of the progeny by the parents). In the brain, this protection would create a window of opportunity for KZFP and TE-driven variation to rewire gene circuits without immediately compromising the host organism. Notably, KZFPs are differentially expressed between brain tissues of human and chimps, two species which protect their progeny during prolonged periods. Together, this suggests a possible role for KZFPs and TEs in the development of higher brain functions specific to modern humans [1].

Breakthrough advances in ancient DNA sequencing methods have culminated in the high-coverage sequencing of two ancient human individuals: a Neanderthal and a Denisovan, yielding unprecedented opportunities to retrace the recent evolution of key genetic features specific to mod-

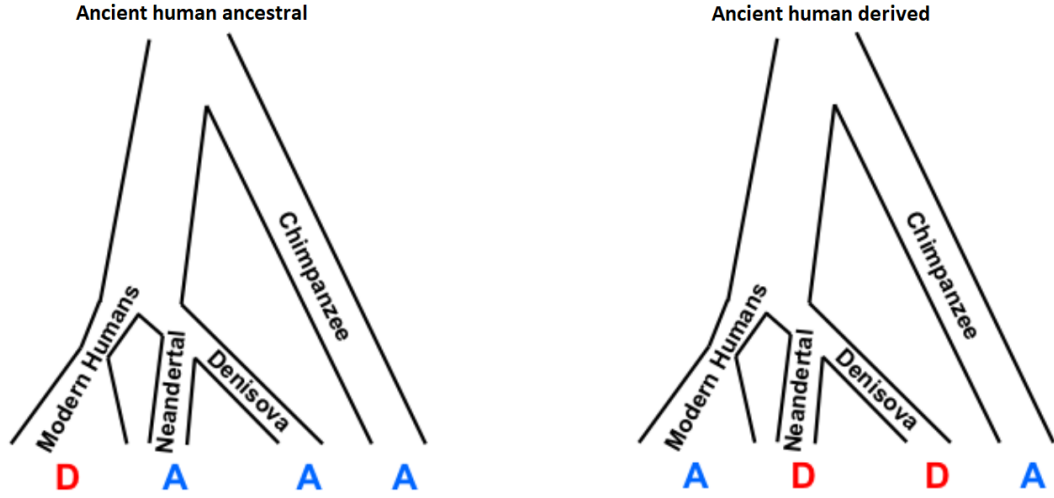


Figure 1: "Ancient human ancestral" alleles (synonymous in the scope of this study to modern human derived alleles) are "chimp-like" in at least one ancient human but different (derived) in more than 90% of the modern human population. "Ancient human derived" alleles are different from the chimp allele in both ancient humans but chimp-like in more than 90% of the human population.

ern humans [5] [4]. While estimates of the population split times of 1) humans and chimpanzees (13 - 6.5 million years ago), 2) modern and ancient humans (550'000 - 765'000 years ago) and 3) Neanderthals and Denisovans (381'000 - 473'000 years ago) vary considerably, the order of these events is well established. Neanderthals and Denisovans belong to a sister group to modern humans, and their genomes provide an invaluable intermediate time point in the evolutionary history separating modern humans from great apes. The depth of coverage of these ancient genomes made it possible to list two categories of single nucleotide changes (SNCs) and small insertions/deletions (inDels) (fig. 1). The first category encompasses SNCs and inDels that were still "chimp-like" (ancestral) in at least one of the ancient humans, but different from the chimp allele (derived) in more than 90% of the human population, suggesting that these alleles have risen to high frequency or fixation after the divergence of modern humans and ancient humans. The second category consists of SNCs and inDels that are ancestral in more than 90% of the human population, but derived both in Neanderthals and Denisovan, suggesting that these alleles have evolved towards strict ancient human specificity after their divergence from the modern human population. Predicting the consequences of both of these collections of DNA variants on the peptide sequence of modern human KZFPs, more specifically on their hallmark functional domains - the KRAB domain and the zinc finger array - represents a direct method to uncover the specificities distinguishing modern human and ancient human KZFP protein families.

In the present work, I address the differences distinguishing modern human and ancient human KZFP families. First, I adapt previously described HMM-based approaches to annotate the human genome for KZFP genes and their corresponding peptide sequences [3]. Second, I derive the putative peptide sequences of eight ancient human specific variants of KZFPs. Third, I show that KZFPs as a family underwent less fixation events both in ancient and modern humans since their

population splitting time than expected for any random gene. Fourth, by comparing functional domains of human KZFPs, I show that C2H2 zinc finger arrays, as opposed to KRAB domains, bear the stigma of a substantial evolutionary pressure aiming at rapidly conferring a unique function. Finally, I trace the recent evolutionary history of ancient human specific KZFPs by comparing the KZFP gene families of humans and apes.

2 Results

2.1 De novo whole genome annotation of KZFP genes

All KZFPs share two obligatory peptide domains: a KRAB domain and a C2H2 zinc finger array. The peptide sequences of these domains are conserved and can be detected using HMM logos. Putative KZFP genes can be subsequently reconstructed using simple orientation, distance and strandedness criterions. I applied this method on human genome assembly hg19 (as Neanderthal and Denisovan genomes were assembled on hg19), which yielded a total of 497 KZFPs, a comparable number to the 489 KZFPs previously detected with a similar method [3]. The lengths obtained for the different domains matched expectations: most KRAB domains span about 120 base pairs (bp), while the few smaller KRAB domains might be 1) degenerated (as suggested by the higher proportion of non protein-coding KZFPs) or 2) split across different exons (fig. 2). The grouping of zinc finger arrays in increments of discrete lengths is characteristic of their multi-unit assembly (multiples of 69 bp segments each coding for a single C2H2, assuming a peptide length of 23 amino acids) and indicates that KZFPs often contain six or more C2H2 zinc fingers. Finally, the spacer domain, defined as the distance between the KRAB domain and the zinc finger array, shows the widest and most unordered DNA length distribution, likely reflecting the reduced selective constraint characteristic of a mostly intronic region. Interestingly, spacer lengths inferior to 500 bp associate with non protein-coding KZFP genes, suggesting a minimum distance required between the KRAB domain and the C2H2 zinc finger array for maintaining protein expression. The number of obtained KZFPs as well as the lengths of their functional domains indicates that the annotation method yields consistent KZFP genes which can be used for downstream analyses.

2.2 In KZFPs, the proportion of canonical C2H2 zinc finger associates with gene type

Previous studies indicate that the DNA-binding sequence of a C2H2 zinc finger is dependent on four DNA-contacting residues in their α -helix. Furthermore, the preferred DNA binding sequence of a C2H2 zinc finger array resembles the concatenation of preferred DNA binding sequences of individual C2H2 zinc fingers [6]. Hereafter, I refer to the concatenation of the groups of four DNA-contacting residues of a KZFP as its "fingerprint". C2H2 zinc fingers were defined as "canonical" when matching a constrained peptide sequence, notably requiring the presence of the two conserved cytosine and histidine residues. I hypothesized that non-canonical C2H2 zinc fingers are non

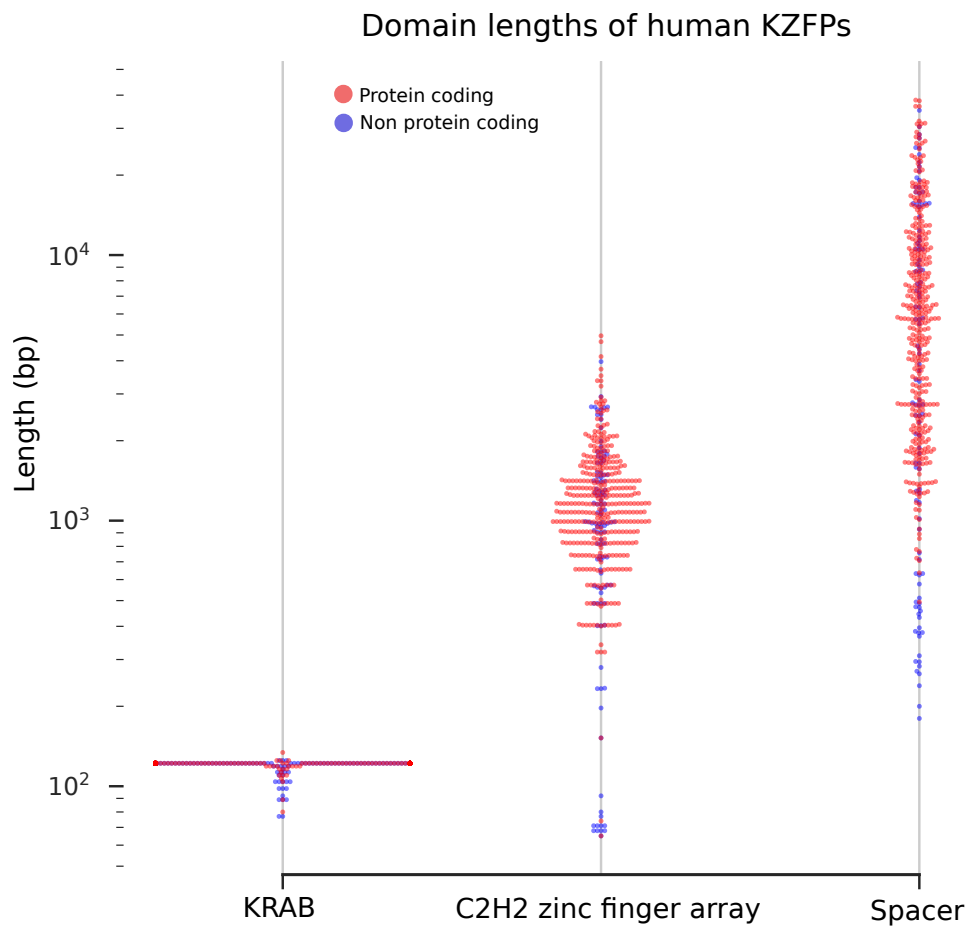


Figure 2: DNA lengths of obligatory domains of KZFPs. The spacer domain is defined as the sequence lying between the KRAB domain and the C2H2 zinc finger array.

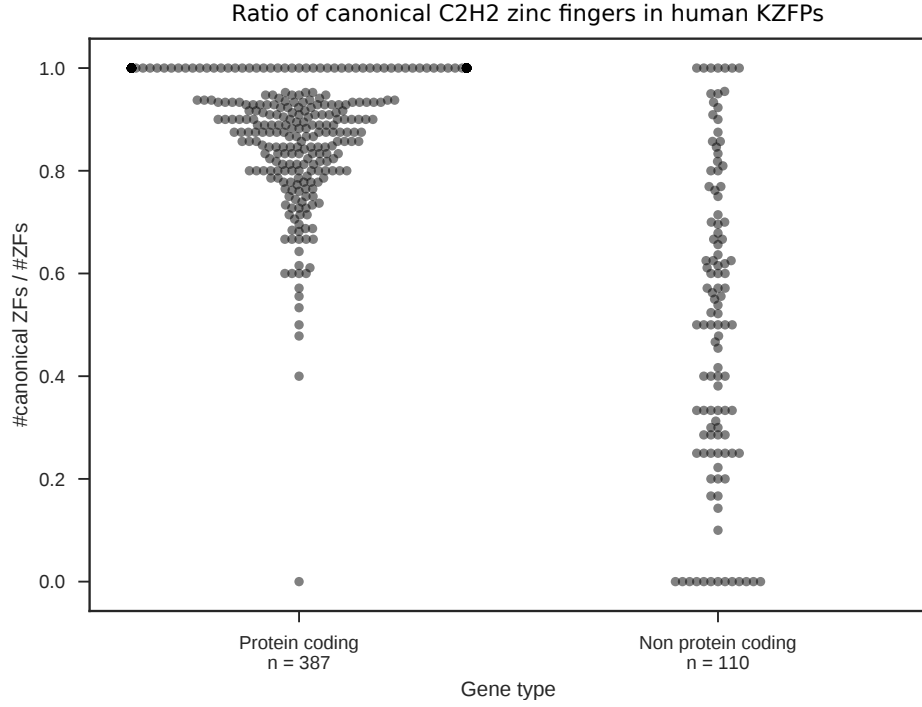


Figure 3: Canonical score for each KZFP. Note that the number of protein coding KZFPs with a canonical score of 1 is 176

functional as they lack the residues required to bind DNA and therefore cannot contribute to fingerprints. The high canonical score (proportion of C2H2 zinc fingers in the zinc finger array) of protein coding KZFPs compared to non-protein coding KZFPs suggests so (fig. 3), as close to half of protein coding KZFPs contain a fully canonical zinc finger array, compared to 6% for non protein coding KZFPs. This reasoning holds for low canonical scores, as no protein coding KZFP has a proportion of canonical C2H2 zinc fingers lower than 0.47. This data suggests that a high proportion of canonical and therefore functional zinc fingers in the C2H2 zinc finger array associates with the gene type (protein coding or not). Note that the two KZFPs with the lowest proportion of canonical zinc fingers are false positives protein-coding genes (RMDN2 and RP11-1396O13.13 are not KZFPs) due to the automatic gene assignment process.

2.3 KZFP genes as a family are conserved between ancient and modern humans

There are two previously published ancient human specific SNCs and inDels collections. The first one contains "ancient human ancestral" alleles (alleles that are still "chimp-like" in either Denisovans or Neanderthals, but different in more than 90% of the current human population). The second collection contains "ancient human derived" alleles (where 90% of the modern human population bears the "chimp-like" allele) [5]. To find out whether KZFP genes are rather conserved or not between ancient humans and modern humans, I asked whether KZFPs as a gene family were

more or less impacted by ancient human SNCs and inDels than expected at random. For both variant collections, I calculated the frequency of SNCs/inDels falling inside specific domains of KZFPs. As a control, I randomly shuffled each domain inside regions restricted to 1) their respective chromosome and 2) exons inside their respective chromosome and calculated the frequency of SNCs/inDels falling inside these.

For both collections of DNA variants (figs 4 and 5, KRAB domains and C2H2 zinc finger arrays are less impacted by SNCs and inDels than spacer domains, reflecting the higher intronic content and decreased conservation of the latter. However, all of the three domains are less impacted by SNCs than expected at random and for exons. The picture is similar for inDels, except for ancient human derived inDels which impacted the spacer more than at random. Together, this data strongly suggests that KZFPs as a protein family underwent few fixation events since the splitting of ancient and modern human populations. This points towards a conservation of KZFPs between ancient and modern humans. Note that the average frequencies of ancient human specific SNCs impacting whole chromosomes and exons are very similar, likely illustrating the small evolutionary distance separating ancient from modern humans.

2.4 Ancient human specific KZFP candidates

To derive the peptide sequences of ancient human specific KZFPs, I evaluated the impact of a collection of previously published SNCs and small inDels on the peptide sequence of human KZFPs. For the KRAB domain and linker regions of the C2H2 zinc finger array as well as for each single C2H2 zinc finger, I considered each of these outcomes for each SNC: synonymous, missense, nonsense or nonstop with respect to the peptide sequence derived from the human reference genome. The few small inDels overlapping with KZFP genes were resolved manually: none of them had an effect on peptide sequences. Only synonymous and non-synonymous SNCs affected KZFPs. Hereafter, I describe the consequences of the SNCs predicted as non-synonymous.

Ancient human ancestral SNCs impacted the peptide sequence of five KZFPs (four protein coding genes: ZNF354C, ZNF763, ZNF726, ZNF221 and one pseudogene: AC023271.1) each time as non-synonymous changes one of their C2H2 zinc fingers. However, a closer examination of individual candidates revealed that three of them were false positives. The non-synonymous SNC impacting ZNF763 falls in a C2H2 zinc finger lying just 3' of a stop codon. The SNC impacting ZNF726 falls inside an out of frame and isolated C2H2 zinc finger lying outside of annotated exons. Finally, the SNC impacting AC023271.1 does change its fingerprint, but the gene product cannot be translated. Indeed its KRAB domain as well as its C2H2 zinc finger array are interrupted by multiple stop codons. This leaves us with two interesting candidates of KZFPs (ZNF354C and ZNF221) impacted by non-synonymous SNCs that probably co-existed at the population level in ancient humans, but rose to high frequency or fixation in modern humans since their splitting with ancient humans. The peptide changes induced by these SNCs do not change the fingerprint nor affect other previously described important residues of the α -helix [6]. Moreover, their relatively low Grantham scores do not predict any obvious functional implications. Interestingly, the

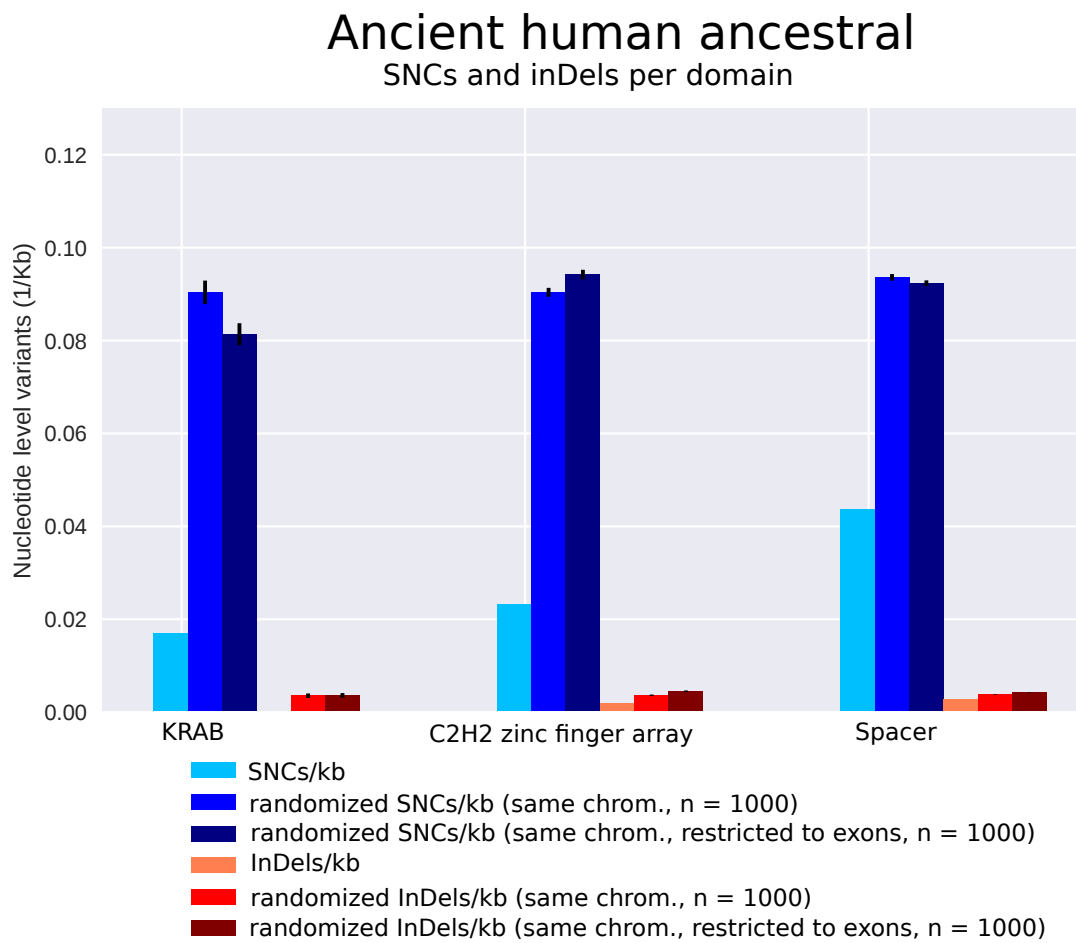


Figure 4: Per domain frequencies of ancient human ancestral DNA variants that became high frequency or fixed derived in modern humans. As controls, windows equal in size to the specified KZFP domains were randomly shuffled across their respective chromosome, or exons on their respective chromosome.

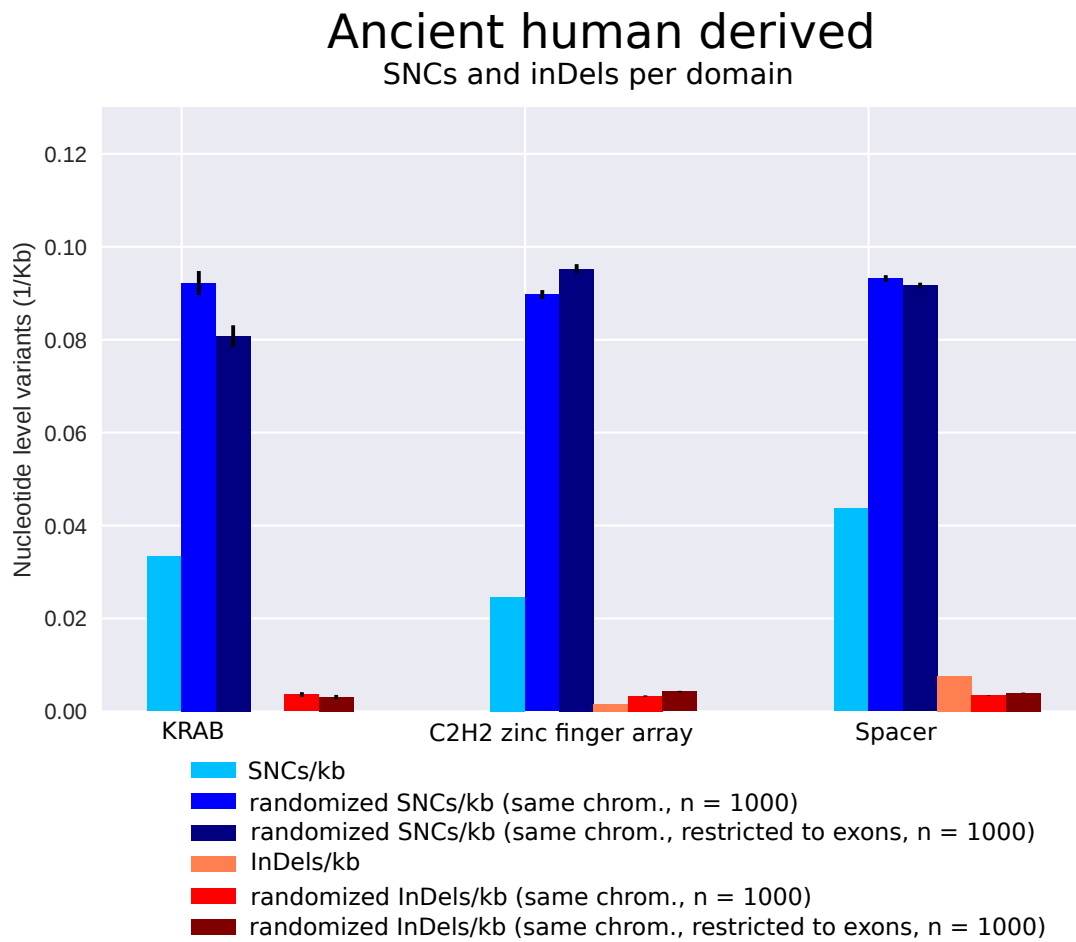


Figure 5: Per domain frequencies of ancient human derived DNA variants that are fixed or high frequency ancestral (chimp-like) in modern humans. Same controls as in fig. 4

SNC impacting ZNF354C was never found in the human population (according to gnomAD[17]), indicative of a truly ancient human specific variant.

Ancient human derived SNCs impacted the peptide sequence of six KZFPs (ZNF487, ZNF138, ZNF616, ZNF112, ZNF619 and ZNF571) all of which are protein-coding. However, two of these hits were likely false positives. Indeed the SNC predicted to affect a linker region in ZNF616 falls more than 1kb 3' of the stop codon reported in all annotated Ensembl exons. The same is true for the SNC affecting a linker in ZNF138 as it lies almost 300 bp 3' of the last stop codon. I examined the specific peptide changes induced by the fixation events in the four remaining KZFPs. The SNC affecting ZNF112 falls inside a coding sequence directly C-terminus of the first C2H2 zinc finger. It was thus predicted as belonging to a linker, although the next C2H2 zinc finger is found more than five peptides downstream of the first C2H2 zinc finger. Therefore the functional implication of this SNC is unclear. The SNC affecting ZNF571 falls inside position -5 of the C2H2 zinc finger α -helix and exchanges a lysine for a glutamic acid, a residue and position combination which has been shown to slightly decrease DNA binding ability of C2H2 zinc fingers [6]. The phenylalanine to serine substitution in the alpha helix of the second zinc finger of ZNF619 stands out by its high Grantham score of 155, reflecting the different properties of these residues (resp. hydrophobic and polar). However I could not find any information concerning the functional role of these residues at this position in the α -helix. Finally, the most extrovert form of ancient human KZFP is ZNF487 which differs at two residues from the modern human variant, one in the KRAB domain and one in the KRAB B box, a peptide sequence that strengthens the recruitment of the transcriptional repressor KAP1 mediated by the KRAB domain [2]. The position bearing the peptide change inside the KRAB domain shows a high conservation score for an alanine according to the HMM Pfam logo [8]. However, the SNC exchanges a proline codon (modern human ZNF487) into an arginine codon (ancient human ZNF487). This is surprising considering that alanine, proline and arginine all belong to distinctive classes of amino acids. The position of the KRAB B box impacted by the second SNC shows a relatively high conservation score for valine, which is replaced by a glycine in ancient humans.

Positively selected regions in the human lineage may be identified by comparing corresponding regions of the ancient human genomes with their modern human counterparts. Positively selected regions are those where the ancient human region is the most dissimilar when compared with corresponding modern human regions from different individuals. In order to detect KZFPs which have been under positive selection since the splitting of ancient and modern humans, I looked for KZFPs overlapping with a previously published collection of such regions. As already reported before [5], two adjacent KZFPs of chromosome 3 (ZNF852, ZNF445) fell inside the smallest of these regions, and are the only genes inside it suggesting that they were positively selected since the splitting of ancient and modern humans. A random shuffling of windows the size of KZFP genes along their respective chromosome followed by intersection with positively selected regions showed that the expected number of intersection is close to 1 (data not shown). Thus finding two positively selected KZFP genes is more than what was expected at random.

2.5 Functional domains of KZFPs bear differing evolutionary pressures

Having established that KZFPs are on average conserved between ancient and modern humans, I wondered whether the modern human versions of ancient human specific KZFPs had features distinguishing them from all other human KZFPs. I addressed this matter in the scope of functional domain homology inside the family of human KZFPs. Because of their role in DNA-binding and assuming an expansion of the KZFP gene family by duplication events, fingerprints have been proposed as indicators of paralogy. This led me to compute the fingerprint similarity (1 for identical fingerprints, 0 when no single residue matched) for each pair of KZFP (fig. 6). The distribution of fingerprint identities is centered below 20% and quickly drops between 40% and 50%, suggesting an evolutionary constraint on fingerprint similarity. Interestingly, more than 40 pairs of human KZFPs - half protein coding and half non-protein coding - share an identical fingerprint, and the distribution does not gradually fall off towards lower values which could simultaneously reflect recent duplication events and/or a rapid reassignment of newly duplicated KZFPs towards unique DNA-binding functions. Examining the maximal identity score for each fingerprint confirmed that 40 unique KZFPs share their fingerprint with at least one other KZFP. Assuming an expansion of the KZFP family by duplication events hints that at least half of these KZFPs are recent compared with all other KZFPs (one duplication creates at least one new fingerprint match). The bulk of the maximum identities being close to 40% indicates that most KZFPs have unique fingerprints and thus probably confer unique DNA-binding capabilities to their corresponding KZFP.

KZFPs impacted by non-synonymous fixation events (both ancient human ancestral and derived) all display a maximum fingerprint identity lower than 60%) meaning they have rather unique fingerprints. However, ZNF852 (positively selected in the modern human lineage) shares more than 90% of its fingerprint with ZKSCAN7, a SCAN-domain containing KZFP gene. This result is in strong contrast with the maximum identity score of ZNF445 (33%) as it is lying in the same positively selected region.

Motivated by the fact that most of the ancient human specific non-synonymous SNCs fell outside of the fingerprint, I compared so-called "negative fingerprints" (residues of C2H2 zinc fingers of the zinc finger array not contained in the fingerprint) across pairs of KZFPs (fig. 7). That the distribution is 1) wider, 2) shifted towards higher identity values and 3) drops equally steeply though at a higher identity than the distribution of fingerprint identities is in agreement with the hypothesis that the negative fingerprint also influences DNA-binding, though being less crucial than the fingerprint for sequence-specificity [6]. Note that the shift towards higher identity values can be partially attributed to highly conserved residues (e.g. cytosines and histidine residues). The high number of pairs of negative fingerprints of low identity specific to non-protein coding pairs likely reflects a higher level of degeneration for otherwise well conserved residues in protein coding pairs.

As for the fingerprint, ZNF852 shares more than 90% of its negative fingerprint with ZKSCAN7. This finding exemplifies that KZFPs sharing a high (more than 90%) fingerprint identity generally also share a high negative fingerprint identity (data not shown).

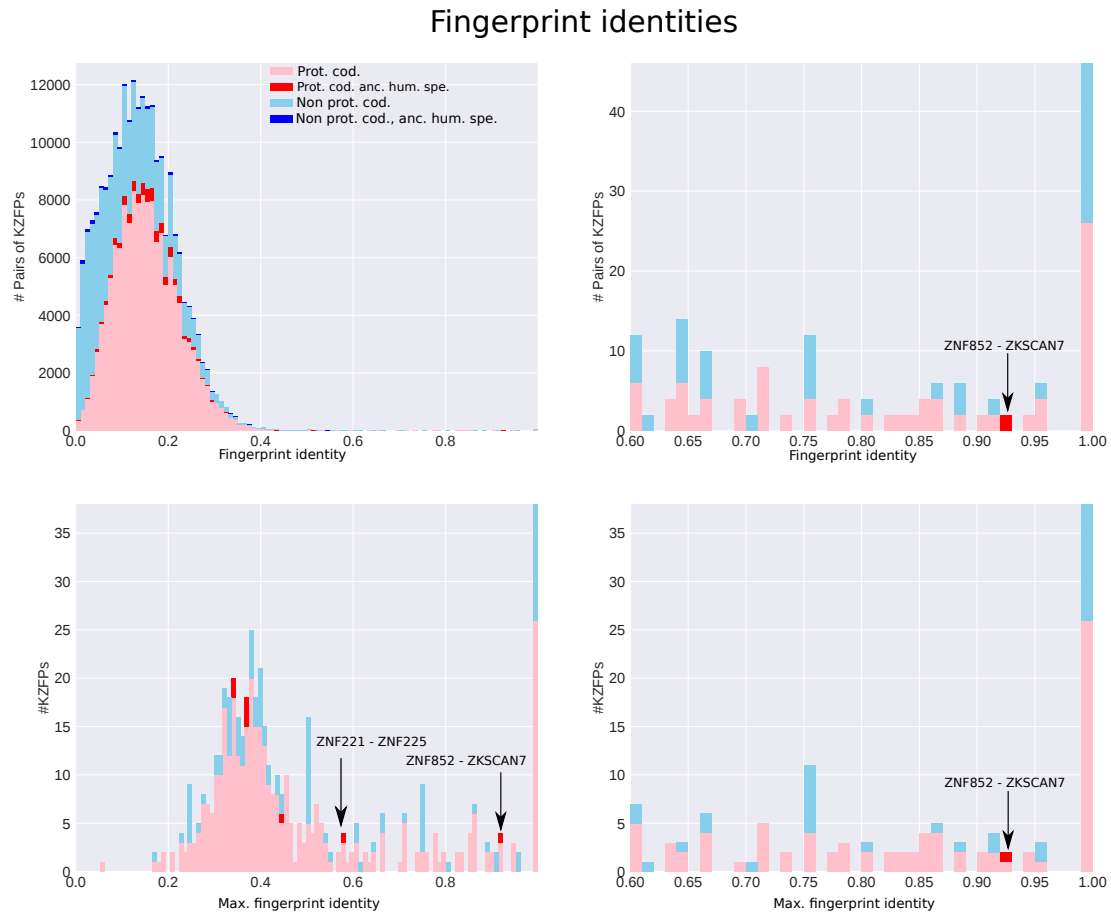


Figure 6: Top: Fingerprint identity (proportion of peptide sequence strict equality) for each pair of human KZFPs. Bottom: maximum fingerprint identity for each KZFP. Protein coding pairs (prot. cod.) contain two protein coding KZFPs. Non protein coding pairs (Non prot. cod) contain at least one non-protein coding KZFPs. Anc. hum. spe. = ancient human specific KZFPs.

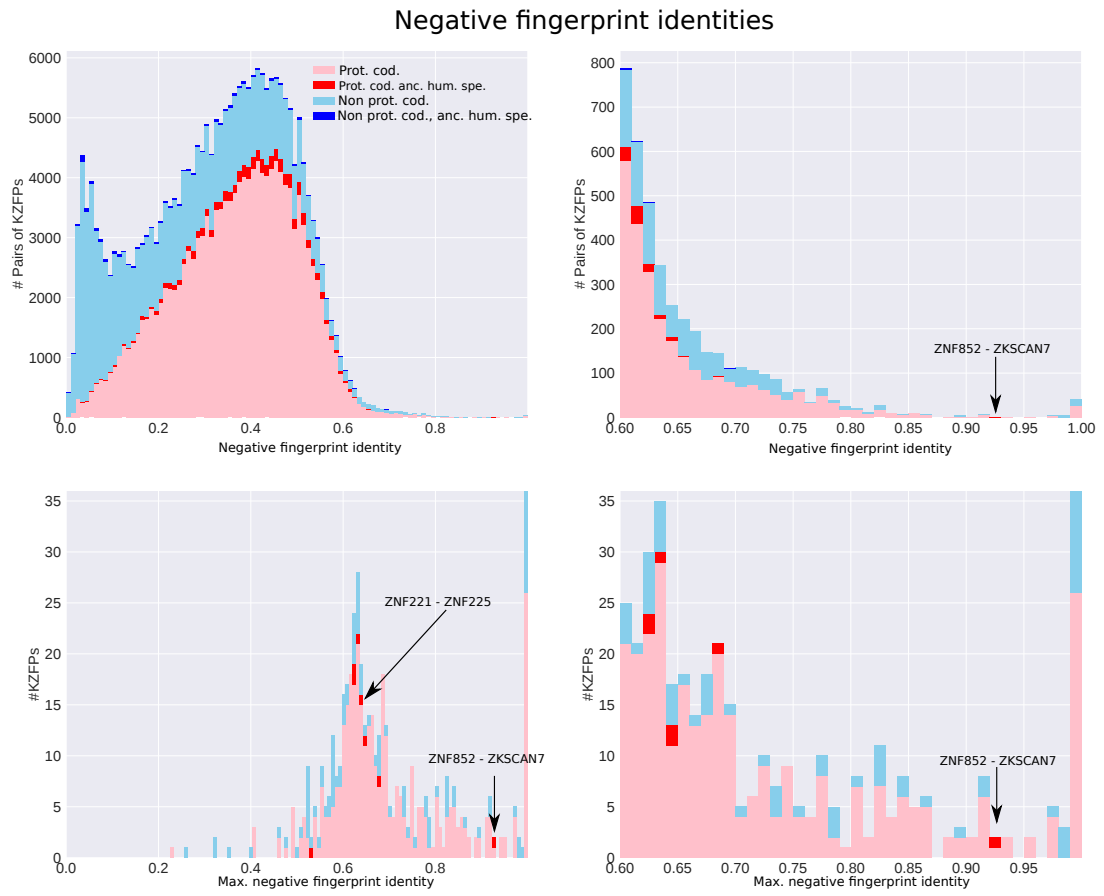


Figure 7: Top: Negative fingerprint identity (proportion of peptide sequence strict equality) for each pair of human KZFPs. Presented as fig. 6.

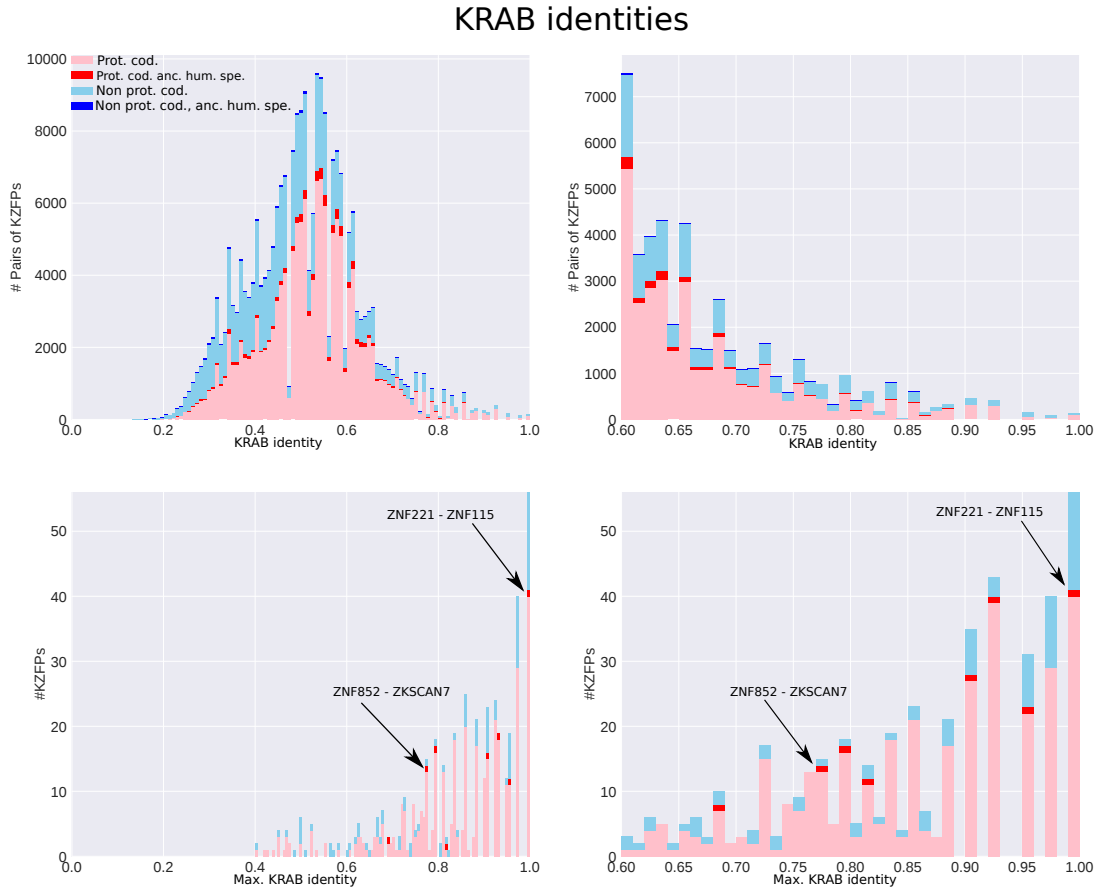


Figure 8: Top: KRAB domain identity (proportion of peptide sequence strict equality) for each pair of human KZFPs. Bottom: maximum KRAB domain identity for each KZFP. Presented as fig. 6.

The distribution of fingerprint and negative fingerprint identities is in stark contrast with that of KRAB domains (fig. 8), the latter being centered near 50% and more symmetrical. The absence of sudden drop in the distribution for high identity values as well as the gradually decreasing distribution of maximal identity scores starting from 100% identity indicate that KRAB domains do not bear the same evolutionary constraint as C2H2 zinc finger arrays.

Surprisingly, ZNF852 and ZKSCAN7, which share a high identity in their C2H2 array, do not share such a high identity for their KRAB domain. Conversely, ZNF221 and ZNF225 display identical KRAB domains but differing C2H2 zinc finger arrays (figs. 6, 7).

The fact that fingerprints (and to a lesser extent, negative fingerprints) display a high degree of uniqueness compared with KRAB domains supports the hypothesis that KZFPs acquire protein-specific functions primarily through modulation of their C2H2 zinc finger array, rather than through modulation of their KRAB domain. However, we previously evoked that an important minority of KZFPs are maintained as protein coding although another protein coding KZFP displays a very similar (if not exactly the same) fingerprint. For these KZFPs, the negative fingerprint was also identical (data not shown). I hypothesized that these KZFPs are maintained as protein-coding due to differing functions carried out by domains other than the C2H2 zinc finger array. Interestingly,

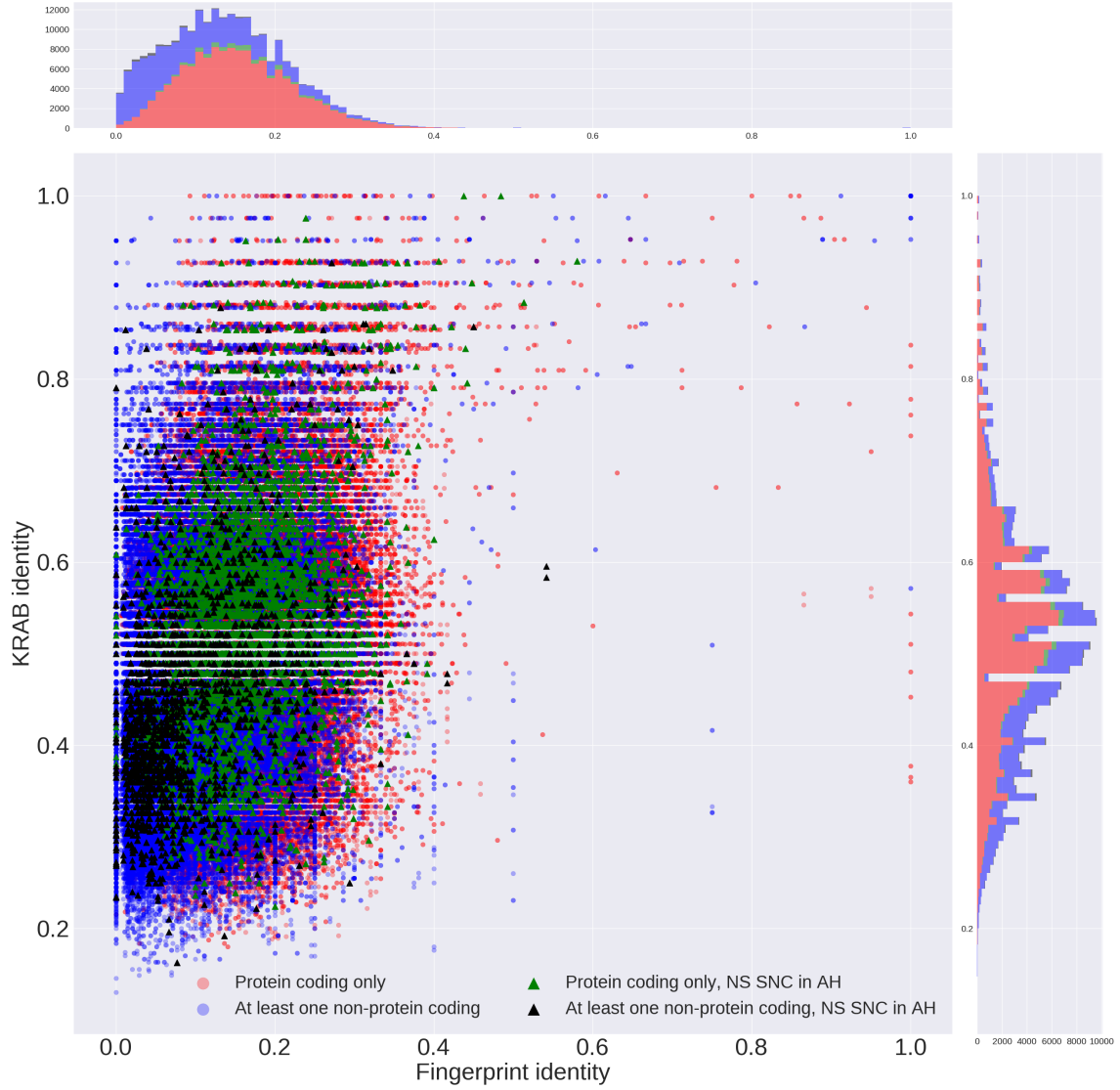


Figure 9: KRAB identity and fingerprint identity for all pairs of KZFPs. NS = non-synonymous, AH = ancient humans

pairs of protein coding KZFPs with an identical fingerprint typically do not have identical KRAB domains (fig. 9), which is not the case for pairs of KZFPs containing at least one non-protein coding KZFP gene. Importantly, pairs of protein coding KZFPs with an identical KRAB typically have dissimilar fingerprints. Taken together, this suggests that although the KRAB domain does not generally evolve to confer function specificity to KZFPs, it may be the case for KZFPs binding to similar DNA-sites.

2.6 Primate history of ancient human specific KZFPs

Human KZFPs were previously dated using a 60% paralogy threshold on the fingerprint [3]. According to this metric, a related form of all ancient human specific KZFPs (including both candidates to positive selection) appeared prior to 75.9 million years ago, the estimated evolutionary distance separating humans from lemurs. However, it was unknown whether more subtle identity

changes occurred in the recent history of apes. To get insight on this matter, I compared each human KZFP to each KZFP detected in the genomes of four decreasingly human-related apes: chimpanzees, gorillas, orangutans and gibbons. I used the mouse as an endpoint and control given 1) the high quality of its genome and 2) the important evolutionary distance separating it from humans, which should translate in a general decrease of KZFP homology.

Changes in the maximum fingerprint identity of each human KZFP compared to non-human KZFPs reflect the evolution of its DNA-binding function (fig. 10). For all three compared domains, the mean identity drops between the gibbon and the mouse, an expected result since the evolutionary distance separating them is much greater than the one separating humans from gibbons. The "saw"-like patterns traced by the identities may either indicate a high level of species-specificity or gaps in the genomes. Ancient human specific KZFPs do not follow specific patterns according to their classes (that is ancient human ancestral, ancient human derived or positively selected), and most of them show an elevated (compared to the mean) and stable level of fingerprint, negative fingerprint and KRAB domain identity throughout the genomes of apes.

ZNF487 stands out from ancient human specific KZFPs as it is maintained at a lower fingerprint identity than other ancient human KZFPs. It is possible that the C2H2 zinc finger array of ZNF487 underwent substantial changes between chimps and humans, but it is equally probable that ZNF487 maintained a high level of variability in the genomes of all apes. It is also unclear whether this peculiarity is linked to the fixation in ancient humans of two non-synonymous SNCs in the KRAB domain of ZNF487.

Surprisingly, the fingerprint identity of ZNF445 (positively selected in the modern human lineage) drops abruptly below 50% in chimps. A closer inspection of the fingerprints revealed that the C2H2 zinc finger array suffered a substantial level of degradation specifically in chimps: it lost several C2H2 zinc fingers while other became non-canonical. This particular case suggests that C2H2 zinc finger arrays of KZFPs may undergo major rearrangements in a relatively small evolutionary window (the evolutionary distance separating chimps from gorillas is close to 3 million years). Note that the KRAB domain of ZNF445 is identical to that of humans in all apes.

3 Discussion

In this study, I aimed at characterizing DNA variants that impacted KZFPs as a gene family in the recent events leading to the speciation of modern humans. I focused on DNA variants which rose to high frequency or fixation either in modern or ancient human populations after both of these split. Two reasons called for a *de novo* human KZFP gene annotation siloed by functional domains (KRAB and C2H2 zinc-finger arrays). First, the highest quality Neanderthal and Denisovan genomes available were assembled by alignment to the human reference genome hg19. This means that DNA variants specific to ancient humans were defined relative to hg19 (and to the variation data from the 1000g project). Second, a collection of human KZFPs with their KRAB and C2H2 zinc finger domains was missing.

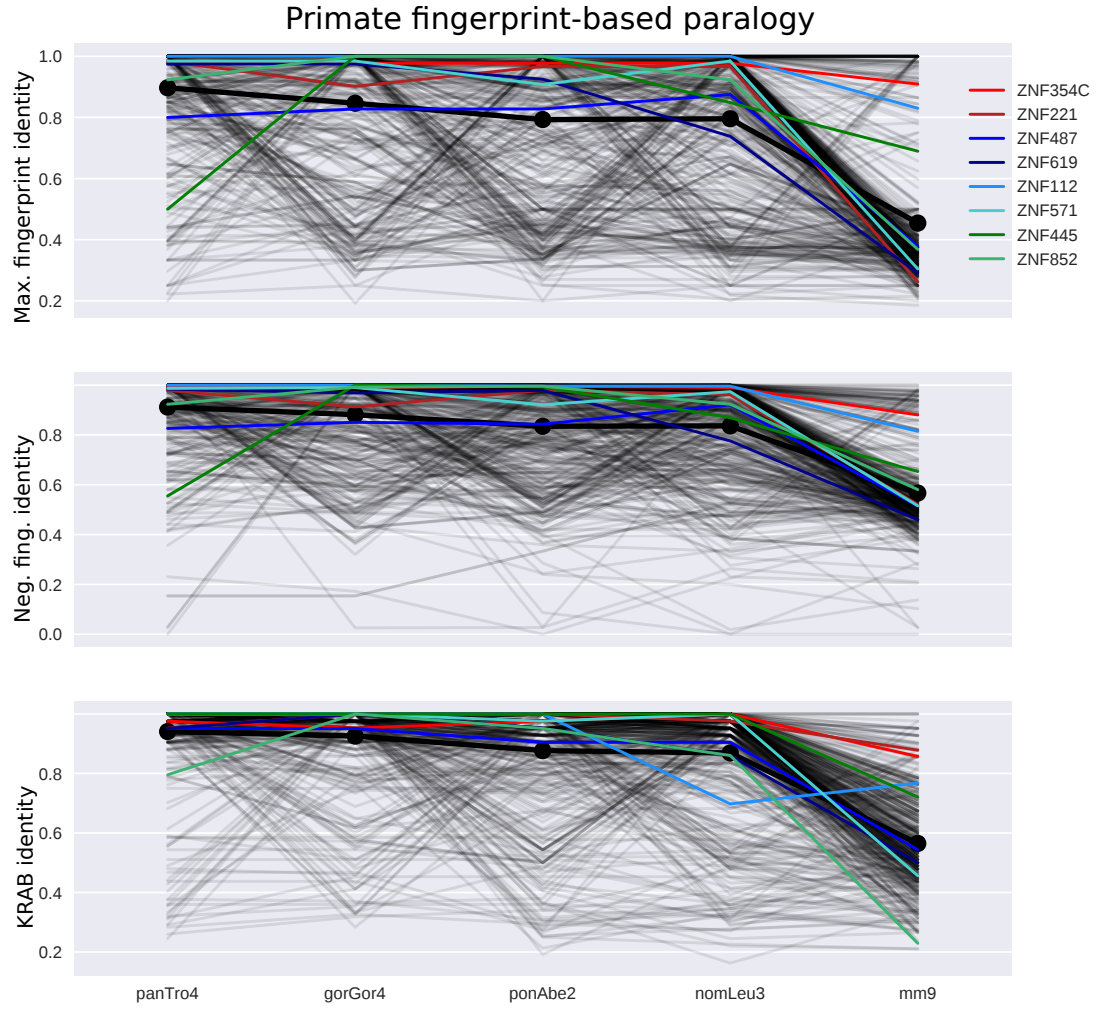


Figure 10: Maximum fingerprint identity for all human KZFPs in the reference genomes of primates. The negative fingerprint and KRAB domain identities are also plotted for the same pairs of KZFPs that yield the maximum fingerprint identity. The thicker black lines with dots represent the means of identity values. panTro4 = chimpanzee, gorGor4 = gorilla, ponAbe2 = orangutan, nomLeu3 = gibbon, mm9 = mouse (used as a control).

While domains are usually very well detected, even in evolutionary remote forms (I (data not shown) and others [3] could extract KZFP genes for the coelacanth using the same pipeline), the method has its weaknesses. First, cases where the KRAB domain coding sequence is separated by intronic content cause either the detection of only one of the two parts of the KRAB domain, or the detection of two KRAB domains instead of one. Second, assigning exactly one KRAB domain and one C2H2 zinc finger array to each KZFP (as opposed to only assigning C2H2 zinc finger arrays as was done previously [3]) leads to a rather complex choice of criteria for KZFP gene assembly, especially on chromosome 19 where KZFP genes lie in close proximity to one another. This notably led to the failed detection of C2H2 zinc finger arrays which are expressed by alternative splicing for ZNF718 and ZSCAN5. However no ancient human specific DNA variants were overlapping these sites. Third, the KRAB domains of ZNF496, ZSCAN25 and ZNF79 were not detected although being well annotated in the literature. It is unclear whether this divergence to the more traditional KRAB domain reflects any functional impact and/or evolutionary peculiarity. In any case, no non-synonymous ancient human DNA variants were impacting these genes. Fourth, some KZFP genes code for additional domains such as the ZSCAN domain, the KRAB B box and the KRAB B div box. I expanded the annotation pipeline on the two latter based on a small set of examples published previously [2], but the low number of seed sequences used to generate the HMM logo might impair a good generalization and thus a good quality of annotation. Expanding the annotation for SCAN domains represents possible improvement to the annotation pipeline. Additional work would be needed to implement the detection of splice sites, 5' UTRs, promoters, full length exons and 3' UTRs as these may also be impacted by fixation events.

Zinc fingers of KZFPs usually belong to the C2H2 family [2] but non-canonical (i.e. probably degenerated) forms of zinc fingers are still readily detected by the annotation pipeline. Thus I wondered whether a high ratio of non-canonical zinc fingers was associated with a maintained functionality, that is a maintained protein-coding state of the gene. Not only was this the case, but there seems to be a threshold close to 50% of canonical zinc fingers to maintain protein-coding functions. Taking the shortcut of protein coding to "useful biological function" suggests a "breaking point" in DNA binding capability when more than half of the zinc fingers are non-canonical. This loss of function could lead to a further degeneration of the ORF (e.g. by accumulation of stop codons). Assuming an expansion of the KZFP family by duplication events, it is tempting to speculate that non-protein coding KZFPs with high canonical scores either lost their protein coding status or inserted in the genome more recently than their counterparts with lower canonical scores as "unemployed" protein-coding DNA sequences should degenerate rapidly.

I then asked how often ancient human DNA variants impacted on the DNA sequence of KZFP genes. All KZFP domains were less impacted by ancient human specific SNVs than expected at random for any position in the genome. This result was expected considering a gene family composed of more than 85% protein coding genes. But surprisingly, all three KZFP domains were less impacted by ancient human specific SNVs than expected for random exons (used as controls for coding regions). This is even more intriguing, first considering that 110 KZFP genes are non

protein coding and second that a substantial proportion of the spacer region, the longest of the three domains (fig. 2), is intronic. This shows that KZFP genes underwent less fixation events both in the ancient and modern human lineages than expected for any random coding region.

The previous observation could either reflect a conservation of fixed alleles or alternatively the maintenance of a high level of variation in KZFP genes of ancient and modern humans. These two cases might even coexist within KZFP genes. To cope with the evolution of TEs and the dynamics of gene networks rewiring, KZFPs should be able to adapt rapidly, for example by switching DNA-contacting residues. In accordance with this model, most fingerprints are unique, and the distribution of fingerprint identities drops abruptly when passing below 1. Furthermore, fingerprints were not impacted by non-synonymous fixation events in either ancient or modern humans. The specific degradation of the fingerprint, but not the KRAB domain, of ZNF445 in chimps, as well as the saw-like patterns formed by human fingerprint identities in the genomes of apes speak for a species-specific and rapid accumulation of variations.

However modularity can only be achieved together with some degree of structural and functional conservation. That all non-synonymous fixation events in the C2H2 zinc finger arrays of either ancient or modern humans occurred in non DNA contacting residues supports this view. Indeed, non DNA-contacting residues of zinc fingers might have an important role for the structure of the zinc fingers themselves, thus enabling a degree of plasticity for DNA-contacting residues. It is even possible that non DNA-contacting residues restrict the range of residues available for DNA contact. In this case, these fixation events could preselect a range of DNA-binding residues to which the KZFP has access. KRAB domains may also be considered as mostly non-modular, as their slowly decreasing maximum identities suggests that they are generally not key to confer function uniqueness. The apparently more constrained structure of KRAB domains makes ZNF487 all the more interesting considering that its KRAB domain displays two non-synonymous SNCs that rose to fixation in ancient humans. Functional studies on the ancient human version of ZNF487 would help to clarify the role of these peptide changes.

In sharp contrast with this general rule, pairs of KZFPs which are maintained as protein-coding despite having identical fingerprints typically display dissimilar KRABs. Conversely, pairs of KZFPs with identical KRAB domains typically do not share similar DNA-binding capabilities. This suggests that in rare cases, the KRAB domain might evolve first and lead to a maintenance of identical C2H2 zinc finger arrays in two distinct KZFPs. It is possible that a difference in other functional domains (such as the SCAN domain) allows the maintenance of highly similar KRAB and C2H2 zinc finger arrays in two or more KZFPs. Note that KZFPs that display similar C2H2 zinc finger arrays could either compete for binding on similar DNA motifs or be expressed separately in a time and cell/tissue specific manner.

ZNF852 and ZNF445 fell inside a putative positively selected region in the modern human lineage. Their relation to other human KZFPs is in stark contrast: ZNF852 has a close C2H2 zinc finger paralog in the human genome (ZKSCAN7), while ZNF445 has a very unique fingerprint. It is unclear whether one of these two KZFPs drove positive selection of the region while the

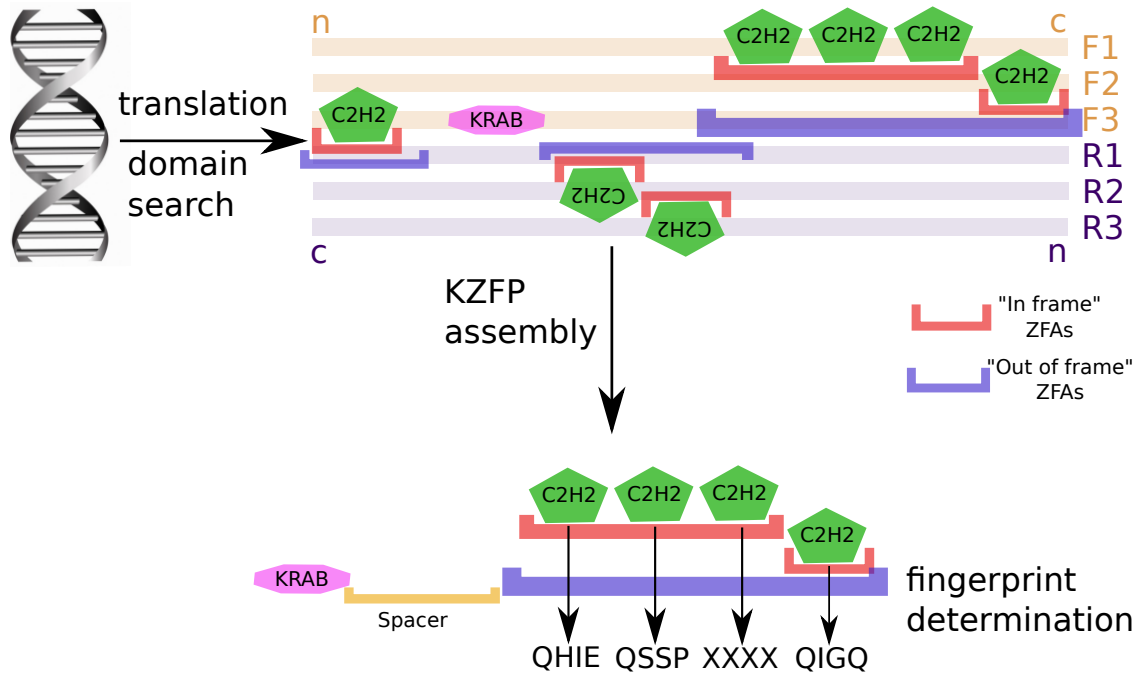


Figure 11: scheme of the annotation pipeline. "XXXX" describes a non-canonical C2H2 zinc finger.

other followed as a bystander, especially since none of these KZFPs seem particularly recent. The coexistence of a ZNF852 and ZKSCAN7 is not primate specific as it can be traced back to the alpaca [3]. However one clue favors ZNF445: its C2H2 zinc finger array underwent a substantial level of change specifically in chimps, whereas it is identical to humans in both gorillas and orangutans. It is possible that ZNF445 was kept under positive selection in the human lineage but not in the chimp lineage, or conversely that it was kept under positive selection in another form in the chimp lineage.

To sum up, KZFPs underwent less SNC/inDel fixation events than expected at random both in the modern and ancient human lineages. This probably reflects a conservation of fixed structures and functions which enables modularity in other domains of KZFPs. Functional studies will be needed to get insights on the consequences of the fixed SNCs that lead to differing peptide sequences in the KZFPs of ancient and modern humans.

4 Methods

4.1 *De novo* whole genome KZFP annotation (see fig. 11)

This section was adapted from [3].

The unmasked hg19/GRCh37 assembly of the human genome was downloaded from UCSC [13]. Chromosomes were six-frame translated using the `translate` function from Biopython [7].

For all non-human genomes, unmapped and alternative chromosomes were included. For the human genome, unmapped and alternative chromosomes were not included.

For each shift in reading frame, an undefined DNA base was added to the chromosome sequence to guarantee that the number of amino acids derived from all translations was equal. For both KRAB and C2H2 zinc finger domains, multiple alignments generated from seed sequences downloaded from Pfam [8] were used to build a Hidden Markov Model (HMM) profile using `hmmbuild` from HMMER2. The HMM profile of KRAB B box was generated by 1) aligning peptide sequences from a previously published list of KRAB B box-containing KZFPs [2] downloaded from UniProt [15] using MUSCLE [16] followed by 2) running `hmmbuild` on a selected peptide region (residues 108 to 124) of the alignment obtained. The HMM profile of divergent KRAB b boxes was obtained in a similar fashion using residues 69 to 102 of the obtained alignment. Next, the HMM profiles were used to scan all whole-chromosome transcripts using `hmmfam` from HMMER2. Single zinc-fingers (sZFs) were selected based on a bit-score threshold of 0, irrespective of stop codons (similarly, 13 for KRAB domains, KRAB B boxes and divergent KRAB b boxes).

C2H2 zinc-finger arrays (ZFAs) were assembled in two steps. First, non-overlapping sZFs sharing the same reading frame and within distance of 1kb were assembled in so-called "in frame ZFAs" (iZFAs). Second, same strand iZFAs within distance of 1kb (overlapping allowed) were assembled in so-called "out of frame ZFAs" (oZFAs).

KZFPs were assembled based on KRAB domains for which a downstream oZFA containing at least two sZFs was found on the same strand, on the condition that the total KZFP gene length (from the first base of the KRAB domain to the last base of the oZFA) was less than 40kb. In the absence of such oZFA, KZFP assembly was tolerated for an oZFA containing only one sZF and satisfying the same gene length and strand conditions. Each KZFP contains a KRAB domain, an oZFA and a so-called "spacer", i.e. the DNA sequence lying in between the KRAB domain and the oZFA. In the case where a KRAB B box or a divergent KRAB B box fitted inside the spacer and respected strandedness, it was assigned to the KZFP. Note that in these cases, the spacer was not redefined.

KZFPs were annotated using the Ensembl annotation for hg19/GRCh37 as downloaded using Biomart [9] [10]. Ensembl genes were kept only if they overlapped at least half of a KZFP gene while being in the same strand using `bedtools intersect` [11]. Each KZFP was assigned to the aforementioned overlapping Ensembl gene with the highest-scoring overlap/(Ensembl gene length) ratio. If no overlapping Ensembl gene was found, the KZFP name was defined as not annotated and considered non-protein coding.

A sZF was defined as canonical when it did not contain any stop codon, and when its amino acid sequence matched the regular expression $\{0,2\}C.\{2,5\}C.\{12,15\}H.\{3,5\}H$, adapted from KRAB domain HMM logo from Pfam. Numbers in brackets specify the minimum and maximum sequential numbers of time the preceding amino acid is allowed to appear in the sequence. A dot represents any amino acid. This regular expression was built based on insertions allowed by the zf-C2H2 HMM profile from Pfam. For a sZF defined as canonical, the fingerprint was defined as previously

described [12]. The first histidine appearing in the regular expression was set as index number 7, and the fingerprint was defined as amino acids indexed -1, 2, 3 and 6 (note that index 0 does not exist). The fingerprint of non-canonical sZFs was set to "XXXX". The negative fingerprint of a sZF refers to all non-DNA-contacting residues, i.e. the amino acid sequence less the fingerprint. For any non-canonical sZF, the negative fingerprint was set to a repeat of "X" of length equal to that of the amino acid sequence minus four.

4.2 Comparison with a reference annotation

ZFAs from a previously published annotation of KZFPs in the human genome [3] (thereafter referred as reference annotation) were converted from GRCh38 coordinates to GRCh37 coordinates using LiftOver from UCSC. Unannotated ZFAs from the RA were named as noname#, (where # represents a unique integer identifier). `bedtools intersect` was used to identify KZFPs generated in my annotation which oZFAs overlap those of KZFPs in the reference annotation. Similarly, KZFPs unique to each annotation (without overlapping oZFAs) were identified.

4.3 Ancient humans KZFP variants

A previously published dataset [5] was used for this analysis. Simply put, the inferred genome from the last common ancestor between Humans and the chimpanzees was defined as ancestral. Any polymorphism (whether single nucleotide change (SNC) or inDel) differing from this reference was defined as derived. Knowing genomic variations at the modern human population level allowed the authors to find alleles which were essentially always derived in the human genome (fixed or at high frequency) but ancestral in at least one of the ancient human genomes. The complementary dataset consisting of high-frequency or fixed ancestral alleles in the human genome but derived in both Denisovan or Neanderthal was also used.

VEP output files were downloaded from http://bioinf.eva.mpg.de/altai_catalog/. Single nucleotide changes (SNCs) present in regions overlapped by KZFPs were extracted. SNCs overlapped by a KRAB domain were resolved in one of the following categories: synonymous, missense, nonsense, or nonstop (see fig. 12). SNCs overlapped by a spacer were not resolved, as spacers were assumed highly intronic. SNCs overlapped by an oZFA but not by any iZFA were not resolved. SNCs present in linkers of iZFA (that is regions not defined as if-C2H2 by `hmmpfam` but linking sZF members of a single iZFA) were resolved as for KRAB domains. Finally, SNCs present in sZFs were resolved as for KRAB domains and linkers, and their impact on the fingerprint was evaluated.

InDels overlapped by KZFPs were extracted and assigned to either the KRAB domain, the spacer or the oZFA. They were resolved manually.

For all non-synonymous changes, the corresponding Grantham score [14] was extracted, and the allelic frequency in the human population was extracted from gnomAD [17].

For all non-synonymous changes in KRAB domains of KZFPs, the reference (i.e. modern human) and the alternative (i.e. ancient human) KRAB domain peptide sequences were aligned to

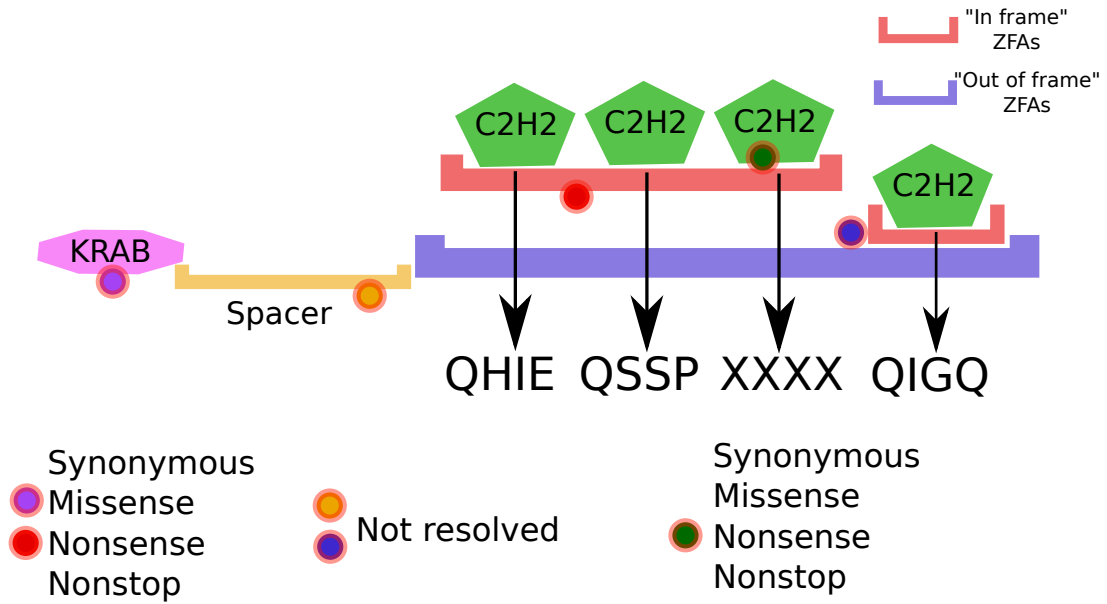


Figure 12: scheme for the resolution of SNCs. "XXXX" describes a non-canonical C2H2 zinc finger.

all other KZFP-assigned KRAB domain peptide sequences using `pairwise2.align` from Biopython, using global alignment and the Blosum80 matrix from `Bio.subsamat.matrixinfo`, complemented with stop codon scores from https://www.ncbi.nlm.nih.gov/IEB/ToolBox/C_DOC/lxr/source/data/BLOSUM80.

Regions inferred to have been affected by positive selection in modern humans since their divergence with ancient humans were extracted from [5] (Table S19a.3). KZFPs were overlapped to positively selected regions in Neanderthal and Denisovan separately using `bedtools intersect`.

4.4 Domain alignments

For each KZFP, fingerprints of oZFA which did not only contain non-canonical sZFs were concatenated as a single sequence, separated by a special character not yet present in the modified Blosum80 matrix described above. Fingerprints of non-canonical sZFs were removed. The matrix was further modified by adding a +100 score when the special character was matched by itself, and a -100 penalty when the special character was matched by any amino acid (but not by a gap). Pairs of KZFP fingerprints were then aligned globally with `pairwise2.align` from Biopython using the modified matrix, an opening gap penalty of -20, a gap extension penalty of 0 and no penalty for end gaps. For each alignment, the score was normalized by the alignment length and corrected for the participation of the special character. The percentage of similarity was calculated as a percentage of matching residues of equal index value. Of note, matching special characters were

removed from the comparison to avoid any bias in calculation of the percentage of similarity, in particular for KZFPs with few sZFs. A similar procedure was employed for the negative fingerprint. The KRAB domain was aligned using usual pairwise alignments settings (global, unmodified BLOSUM80 matrix, opening and extending gap penalty of -1, penalizing end gaps).

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